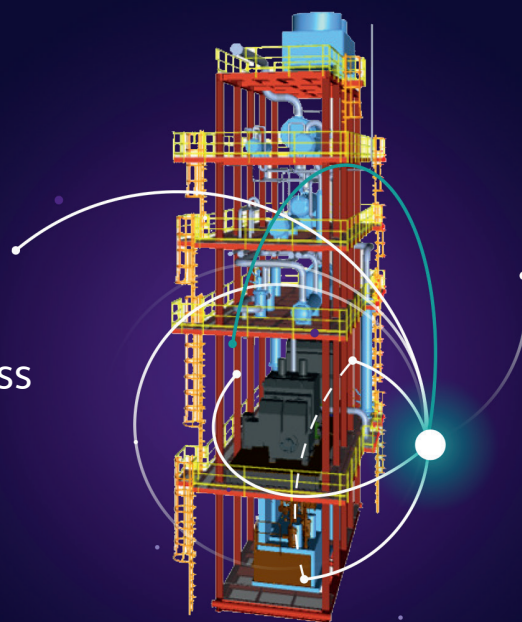


Rotating Olefins Cracker (ROC)

A novel, game-changing approach to decarbonize the olefin production process

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The Rotating Olefins Cracker (ROC), jointly developed by Siemens Energy and Technip Energies, introduces a brand-new approach to olefins production.

It replaces the conventional cracking furnace with a novel type of turbomachine, capable of producing the high temperatures required for the pyrolysis reactions directly within the process fluid, through the process of a shock wave. The heating is very rapid and applied directly to the feedstock without heat transfer.

The ROC turbomachine supports driver flexibility, and when driven by electric-powered motors or hydrogen-fired gas turbines, this technology unlocks multiple pathways to decarbonize the process used to produce light olefins.

Key benefits:

- **Lower GHG Emissions**
Rotating machinery can utilize carbon-free energy (e-drive or H2 GT drive); potentially eliminating all CO2 emissions for a fully electrified plant
- **Yield**
 - Near-instant heating
 - Ultra low residence time
 - Better cracking selectivity
- **Coking**
 - No surface heat exchange (no radiant coils)
 - Minimum potential to form carbon deposits
- **Size**
Compact turbomachinery, heat input not constrained by exchanger surface area
- **Scale-up**
Technology inherently capable of high throughput, 2x flow << 2x size, uses proven range of electric motor drivers for commercial scale units.

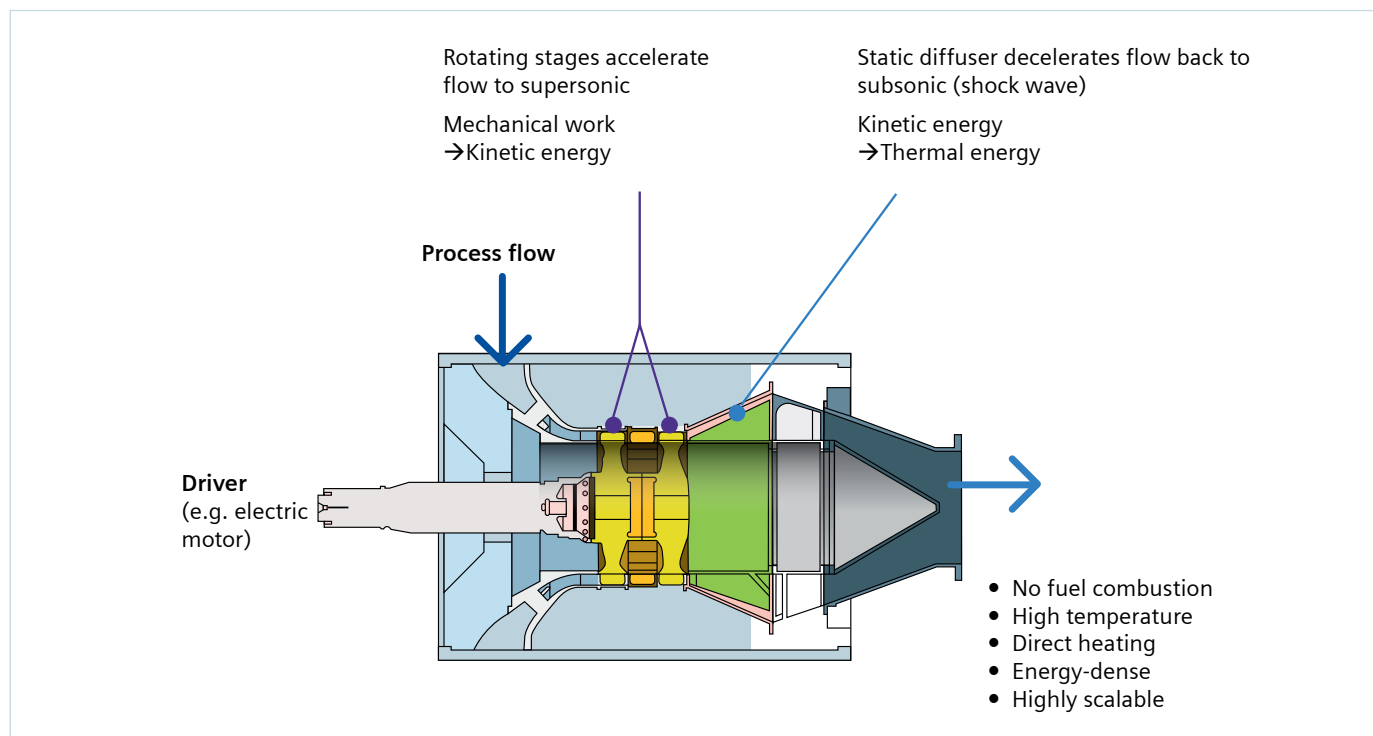
Modular design:

The ROC machinery is much smaller than the radiant section of a conventional furnace, allowing the unit to be much more compact. It can be supplied in modules and adapted to various site footprints.

The quench and feed heating exchangers are conventional but can be replaced by Technip Energies' patented gas/gas primary quench exchanger to reduce the electricity and steam consumption in the feed and dilution steam heating duty by 20-25%. Finally, the turbomachinery based solution scales up well, doubling output for much less than double the equipment size.

How it works:

Shock wave heating turbomachinery



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